

Inverse Airfoil Design as Constrained Root-Finding: A Monolithic CST–Newton Formulation

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Abstract

We recast inverse airfoil design – recovering a geometry that produces a prescribed surface pressure or edge-velocity distribution – as a single determined nonlinear root-find rather than an objective-function search. Parameterising the surface with Class-Shape-Transformation (CST) coefficients that enter the geometry *linearly* makes the geometric Jacobian block exact, constant, and design-independent, and makes geometric design constraints (leading-edge radius, trailing-edge thickness, inscribed area) linear algebraic rows rather than nonlinear predicates. Appending the CST coefficients as unknowns to a coupled viscous/inviscid Newton solver (mfoil) therefore converts constrained shape *optimization* into constrained *root-finding*: one square system, no outer loop, no surrogate. We validate the architecture with a falsifiable self-consistency test – recovering a known CST coefficient vector from its own self-generated target – to $\|A - A^*\|_\infty = 1.079 \times 10^{-11}$ in 4 Newton iterations with a visible quadratic tail, and confirm the recovered geometry reproduces the target in fully released (natural-transition) direct mode to $\Delta c_l = 8.0 \times 10^{-13}$. An ablation matrix identifies the primary uniqueness guard for the resulting square system: sensitivity-optimal (QR-pivoted) selection of target stations, not initial-guess quality, is what separates recovering the true design from converging cleanly to a spurious but residual-zeroing root. Measured against a competently-tuned nested Levenberg–Marquardt baseline under two independent fair-paired controls, the monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations – a real but modest reduction, not the two-to-three-orders-of-magnitude headline hypothesised a priori – while retaining determinism, guaranteed local quadratic convergence, exact analytic constraint-row Jacobians, and per-iteration failure-mode diagnostics that the nested baseline has no analogue for.

1 Introduction

Inverse aerodynamic design – given a target surface pressure or isentropic Mach distribution, recover the geometry that produces it – is classically posed either as a well-posed but restrictive conformal-mapping problem [Lighthill, 1945, Volpe and Melnik, 1986], or as an outer optimization loop wrapped around a converged forward flow solve. The latter is the practical default in industry: a geometry parameterisation, a surrogate or gradient-based optimizer, and hundreds to thousands of fully converged flow solves per design. The present paper’s motivation is autobiographical: the author’s own doctoral work on sCO₂ axial-turbine blading used exactly this pattern – Class-Shape-Transformation (CST) geometry, a B-spline control-point wrapper, and a Dakota/Kriging surrogate loop against MISES – and its future work section named the open problem verbatim: integrating CST directly with a Newton-based flow solver to escape the nested loop (docs/CST_MISES_Monolithic_Inverse_Design.md §1.4).

That escape route already exists in one solver. MISES’s Modal-Inverse mode [Drela, 1990] appends mode-shape amplitudes directly to its global Newton system and drives them so that

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computed surface speed matches a target – a determined root-find, not a search. This is not new; Drela described it in 1986–1990. What MISES’s built-in mode shapes are *not*, however, is a complete, geometrically meaningful, analytic design basis: they are heuristic bump functions, not a family whose coefficients double as leading-edge radius, trailing-edge thickness, or inscribed area. Kulfan’s CST parameterisation [Kulfan, 2008] is such a basis, and Masters et al. [Masters et al., 2017] showed it is among the most efficient analytic families across a 2000-airfoil comparison. CST has also already been coupled to inverse design once, by Morris, Allen & Rendall [Morris et al., 2010]: there, CST is used to *smooth* a pressure-residual-driven normal-direction shape correction inside an outer iterative loop – CST is the smoother, not an unknown in a Newton system. The present work does not repeat either of these; it combines a property of CST that neither prior use exploits.

The precise novelty claim. CST’s surface parameterisation

$$\zeta(\psi) = C(\psi) \sum_i A_i S_i(\psi) + \psi \zeta_T \quad (1)$$

is *linear* in the coefficient vector A . Two consequences follow, and neither is available to a generic (e.g. B-spline-with-wrapper) geometry representation once local-frame perturbations, LE tangent-angle displacements, or re-fitting are introduced to reach the same design freedom (docs/CST_MISES_Monolithic_ (1) the geometric Jacobian block $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ is exact, constant, and design-independent – assembled once, never re-derived inside a Newton loop; and (2) the geometric functionals that matter for manufacturability (LE radius, TE thickness, wedge angle, inscribed area, even leading-order curvature continuity) are *linear* in A , so they enter the Newton system as constant-coefficient algebraic rows rather than non-smooth predicates or penalty terms. The contribution of this paper is **not** “geometry unknowns in a Newton system” (that is Drela, 1986/1990) and **not** “CST used somewhere inside inverse design” (that is Morris, Allen & Rendall, 2010). It is the combination: an analytic basis whose linearity makes the geometric Jacobian exact and constant, *and* whose geometric functionals make design constraints linear rows, together converting *constrained shape optimization* into *constrained root-finding* – one square Newton system, solved once, no outer loop.

The honest headline. A priori, the architecture was hypothesised to require “two to three orders of magnitude fewer flow solves, plus determinism and uniqueness” relative to a nested-optimization baseline (docs/CST_MISES_Monolithic_Inverse_Design.md §1.3). That number was never measured against a competent baseline; it is not repeated here. What we report instead, from a controlled ablation study (§5) against a modern, properly-scaled, tightly-tolerance Levenberg–Marquardt solver [Virtanen et al., 2020]: **3–8x fewer flow-residual evaluations, depending on initialization strategy**, under two independent fair-paired controls (paper/results_digest.md §5.3), plus qualitative properties the flow-solve ratio does not capture – determinism (no stochastic restarts), guaranteed local quadratic convergence, exact analytic constraint-row Jacobians (no finite-difference noise in the constraint block), and a battery of per-iteration failure-mode diagnostics (§3) the nested baseline has no equivalent of. We regard the identification and correction of the a priori $\geq 100x$ claim, together with the discovery of a station-selection identifiability mechanism that the original hypothesis did not anticipate (§5.1), as evidence that the falsifiable-test discipline (§4) this paper follows throughout is doing real epistemic work, not merely validating a foregone conclusion.

The remainder of the paper: §2 states the CST mathematics, the constraint rows, the extended Newton system and its degree- of-freedom accounting, the derivative strategy actually forced by

the target solver’s exposed interface, and the two solver-specific closures (forced transition, pre-solve/realisability) needed to make the Newton iteration well-posed. §3 catalogues the four failure modes the architecture is vulnerable to and the diagnostics built to discriminate between them. §4 describes the self-consistency test that falsifies or confirms the hypothesis outright. §5 reports the T8 ablation matrix: the identifiability finding, the Bernstein-conditioning cliff, the corrected flow-solve comparison, and the quadratic-convergence evidence. §6 places the four solver-specific mitigations in a common TRIZ separation-principle frame, states the study’s honest scope limits, and discusses the outdated-baseline finding. §7 closes with the Stage 2 (cascade) and Stage 3 (MISES) extensions this architecture was designed to scale into.

2 Formulation

2.1 CST parameterization and its linearity

Let $\psi = x/c \in [0, 1]$ and $\zeta = z/c$. The CST class function for a round-nose, sharp-tail family is

$$C_{N_1}^{N_2}(\psi) = \psi^{N_1}(1 - \psi)^{N_2}, \quad N_1 = 0.5, \ N_2 = 1.0, \quad (2)$$

and the shape function is a Bernstein-polynomial expansion of order n per surface,

$$S_i(\psi) = K_i \psi^i (1 - \psi)^{n-i}, \quad K_i = \binom{n}{i}, \quad S_u(\psi) = \sum_{i=0}^n A_{u,i} S_i(\psi), \quad S_l(\psi) = \sum_{i=0}^n A_{l,i} S_i(\psi). \quad (3)$$

The upper and lower surfaces are

$$\zeta_u(\psi) = C(\psi) S_u(\psi) + \psi \zeta_{T,u}, \quad \zeta_l(\psi) = C(\psi) S_l(\psi) + \psi \zeta_{T,l}, \quad (4)$$

with $\zeta_{T,u} = +t_{TE}/2$, $\zeta_{T,l} = -t_{TE}/2$ (sign convention binding in `src/cins/CLAUDE.md`; `SPEC.md` §2.1).

The property this paper exploits is that Eq. (4) is *linear* in A : the partial derivative

$$\frac{\partial \zeta}{\partial A_i} = C(\psi) S_i(\psi) \quad (5)$$

does not depend on A . It is a fixed matrix, computed once from the chordwise node distribution and cached (`SPEC.md` §2.1; T2 requirement, `docs/CST_MISES_Monolithic_Inverse_Design.md` §7.3): *“does not depend on A. Recomputing it per Newton iteration is the single most common way to lose the performance argument.”* This design-independence is what makes the $\partial R/\partial A$ Jacobian block of §2.3 assemble as a cheap matrix product rather than a re-derived, geometry- dependent operator at every Newton step.

Two endpoint identities follow directly from Eq. (4) and are used as constraint rows in §2.2:

$$A_0 = \sqrt{2R_{LE}/c}, \quad A_n = \tan \beta + \Delta z_{TE}/c, \quad (6)$$

where β is the boat-tail (trailing-edge wedge) angle. Near $\psi \rightarrow 0$, $\zeta \approx A_0 \sqrt{\psi}$, so $R_{LE} = A_0^2/2$; G^2 curvature continuity across the leading edge reduces to matching the leading coefficient magnitude on the two surfaces, $A_{u,0} = |A_{l,0}|$ – under the stored-sign convention ($A_{l,i} < 0$), the row $A_{u,0} + A_{l,0} = 0$ (`SPEC.md` §2.3).

2.2 Constraints as linear rows

Because Eq. (4) is linear in A , every geometric functional that is itself linear or integral in ζ becomes a linear row $g \cdot A = b$ rather than a nonlinear predicate evaluated on a rebuilt mesh. The leading-edge-radius and trailing-edge-wedge rows follow immediately from Eq. (6). The one requiring a closed-form integral is inscribed area: the per-surface contribution is

$$\int_0^1 C(\psi) S_i(\psi) d\psi = K_i \int_0^1 \psi^{i+N_1} (1-\psi)^{n-i+N_2} d\psi = K_i B(i+N_1+1, n-i+N_2+1), \quad (7)$$

a Beta function, plus the trailing-edge term $\int_0^1 \psi \zeta_T d\psi = \zeta_T/2$. Eq. (7) is evaluated with `scipy.special.beta` – never by numerical quadrature in a production constraint row; quadrature is retained only as an independent gate check (SPEC.md §9, “Do not”). That gate passed: `area_row` matches numerical quadrature of the fitted NACA 2412 to 10^{-10} (docs/gates/T3-closure.md; SPEC.md §7, T3 row). This closed-form machinery is the same Beta-function apparatus Joseph & Mohan [Joseph and Mohan, 2021] use for thin-airfoil lift/moment integrals against a Bernstein camber basis – not a coincidence, since both integrate Bernstein polynomials against the class function, and it is the analytic kernel reused for the linear pre-solve of §2.6.

Under a B-spline-with-wrapper representation, by contrast, every row in Table 1 is a nonlinear or non-smooth predicate evaluated on a rebuilt mesh – which is precisely the stack that forced surrogate-based search in the author’s PhD work (docs/CST_MISES_Monolithic_Inverse_Design.md §1.5).

Table 1: Constraint representation: CST vs. a B-spline control-point wrapper. Source: docs/CST_MISES_Monolithic_Inverse_Design.md §1.5.

Constraint	Under CST	Under the B-spline wrapper
LE radius	$A_0 = \sqrt{2R_{LE}/c}$ – linear, exact	implicit, via tangent points
TE thickness	ζ_T term – linear	geometric post-check
Boat-tail/wedge angle	A_n – linear	nonlinear angular DV
Inscribed area	linear functional, Beta-function closed form	numerical, non-smooth
Curvature	rational function of A , differentiable	monotonicity predicate, non-smooth

2.3 The extended Newton system and DOF accounting

The extended system appends the CST coefficients (and, optionally, angle of attack) directly to the flow solver’s own global unknown vector. Unknowns:

$$U \text{ (flow state, } N_{tot}), \quad A \text{ (CST coefficients, } n_A), \quad \alpha \text{ (angle of attack, 0 or 1)}. \quad (8)$$

Equations:

$$R(U; x(A)) = 0 \text{ } (N_{tot}), \quad T(U) - C_{p_{target}} = 0 \text{ } (M), \quad G \cdot A - b = 0 \text{ } (K), \quad (9)$$

with the block Jacobian

$$J = \begin{bmatrix} \partial R/\partial U & \partial R/\partial A & \partial R/\partial \alpha \\ \partial T/\partial U & 0 & \partial T/\partial \alpha \\ 0 & \partial G/\partial A & 0 \end{bmatrix}. \quad (10)$$

$\partial R/\partial U$ and $\partial R/\partial \alpha$ are mfoil’s own analytic, sparsely-assembled blocks, reused unchanged. $\partial T/\partial U$ is a near-selection matrix (Cp at a node as a direct function of the panel/edge-velocity unknowns).

$\partial G/\partial A = G$ is constant – free, per §2.2. $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ is the block that carries the architectural claim; its second factor is the cached, design-independent matrix of Eq. (5), and its first factor is addressed in §2.4.

DOF accounting. The dossier’s original bookkeeping (docs/CST_MISES_Monolithic_Inverse_Design.md §4 F1, §7.6) is

$$\text{DOF} = 2(n + 1) - 1_{\text{shared LE}} - 1_{\text{TE fixed}} + 3_{\text{scale/rotate/translate}} + 1_{\text{stagger or } \alpha} \stackrel{!}{=} M + K.$$

In the Stage-1 (isolated-airfoil, mfoil) implementation, the three rigid overall modes (scale/rotate/translate) are not exposed as free Newton unknowns, so the implemented squareness condition is the simpler, equivalent statement actually asserted at construction time and logged every run (SPEC.md §3):

$$M + K = n_{A,\text{free}} + n_{\alpha}, \quad (11)$$

where $n_{A,\text{free}}$ is the count of CST coefficients *not* eliminated by a prescribed-LE treatment (§4) or an equality-row elimination, and $n_{\alpha} \in \{0, 1\}$ depending on whether angle of attack is a free unknown for the run in question. Violating Eq. (11) raises at construction, before any Newton step – this is the FM-1 guard (§3), and it is exercised directly in the T8 ablation matrix (§5.1): deliberately over- and under- squaring the baseline system ($n=8/\text{side}$, $n_{A,\text{free}} = 16$, $n_{\alpha} = 0$) produces the exact, actionable diagnostics

```
"extended system not square: M+K=17 but n_A_free+n_alpha=16 (M=17, K=0, n_A_free=16)"
"extended system not square: M+K=15 but n_A_free+n_alpha=16 (M=15, K=0, n_A_free=16)"
```

with zero Newton iterations spent and no silent wrong answer (paper/results_digest.md §5.2 “Other single-factor ablations”; experiments/results/t8/ANALYSIS.md §4). Getting Eq. (11) wrong is, per the dossier, plausibly what the author’s PhD-era divergence was misdiagnosed as an LE-singularity problem when it was rank deficiency (docs/CST_MISES_Monolithic_Inverse_Design.md §4, FM-1) – instrumenting for it directly, rather than inferring it from Newton behaviour, is the point of Eq. (11) being an assertion rather than a derived quantity.

2.4 Derivative strategy: why finite-difference-over- A , not the analytic chain rule or complex-step

SPEC.md’s derivative-mode decision tree (§5) prefers, in order: (1) the analytic chain rule $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ if mfoil exposes $\partial R/\partial x$ analytically; (2) complex-step over A if the residual-assembly path is free of **abs**/**max**/**min**/comparison operators; (3) real (central) finite differences over A otherwise. T1 introspection (docs/mfoil_internals.md §3–4) resolved both questions empirically, and neither resolved in the analytic-chain-rule’s favour:

- **The analytic chain rule is incomplete for this solver class.** mfoil exposes `M.glob.R_x`, but it is only a *partial* sensitivity – real and useful for the arc-length (ξ) mapping, but missing the geometry-dependent operators that matter most: the inviscid panel/AIC coupling and the stagnation-point formulation (docs/mfoil_internals.md §3, “KEY QUESTION A”). Using this partial block as if it were the complete $\partial R/\partial x$ would silently drop exactly the rows the leading-edge failure mode (FM-3, §3) is about.
- **Complex-step is blocked, not merely inconvenient.** A bare `residual_station/residual_transition` call, with fixed panels and a complex-perturbed state, is verified complex-step-safe and reproduces the analytic R_x to float64 machine epsilon (docs/mfoil_internals.md §4.1). But the *full*

augmented residual – which is what a perturbation in A actually varies, since A moves the panel geometry itself – calls the panel-influence and stagnation-detection code paths, which use `atan2`, branch selection on real quantities, and a hard `ValueError` guard for complex node arrays (docs/mfoil_internals.md §4.1–4.2, `mfoil.py:2713`). These are exactly the operators complex-step cannot tolerate, and they sit in precisely the geometry-dependent block that (1) shows was missing from the analytic path – i.e. complex-step *cannot fill the gap the analytic chain rule leaves* (docs/mfoil_internals.md §8, “picture §3.2. Complex-step cannot fill that gap either”).

The implemented and gate-verified strategy is therefore real central finite differences with respect to A only – never with respect to node coordinates x (SPEC.md §5, “Hard rule (never violate)”; SPEC.md §9). This is a deliberate architectural discipline, not a fallback of convenience: A has $n_A \approx 16\text{--}36$ columns depending on Bernstein order, so a full central-difference Jacobian block costs $O(n_A)$ residual *evaluations* per Newton step – not flow solves – against the alternative of finite-differencing over $N_{\text{panel}} \gg n_A$ node coordinates, which would silently destroy the flow-solve-count argument this paper’s Results section depends on. T5 gate closure verified the resulting Jacobian columns to 5–6 significant figures across a step-size sweep with no stencil discontinuities (docs/gates/T4-T7-closure.md, “T5”).

2.5 Forced-transition onset closure

mfoil’s (and XFOIL’s) e^n transition closure is only C^0 in the design variables: the transition-station equation itself performs an internal sub-Newton solve for the point x_t where the marched amplification factor crosses n_{crit} , and that location moves discontinuously as geometry changes cross a threshold. A coupled Newton iteration needs a continuous residual; a kink stalls or chatters it (dossier §4, FM-4). The mitigation prescribed by the dossier is to fix (trip) transition during the inverse iterations and release it for a verification solve afterward. mfoil exposes no forced-trip API, so this is implemented as an adapter-level shim (ADR-0003) rather than a vendor-code edit.

The first implementation attempt froze the laminar/turbulent node flags (`M.vsol.turb`) and no-op’d mfoil’s own `update_transition`, but left the natural `residual_transition` equation untouched at the frozen boundary station. This was *insufficient*: empirically (NACA 2412, $\alpha = 2^\circ$, $Re = 10^6$, trip 5%/5%), the physics moved correctly under Newton pressure (c_d : 0.00578 $\rightarrow$ 0.00741) but the solve never converged – `Rnorm` stalled at ≈ 12.6 for the full 50-iteration budget, because the frozen station equation still solves internally for amplification = n_{crit} , which has no solution once transition is pinned somewhere amplification is nowhere near n_{crit} (ADR-0003, “Revision”). The corrected mechanism, `_residual_transition_forced`, replaces the boundary-station residual with an XFOIL-style turbulence-*onset* closure: the momentum and shape-parameter rows use the ordinary turbulent `residual_station` closure directly across the whole station (no laminar sub-leg, no station-local x_t), and the shear-lag row is replaced with the same onset condition mfoil’s own `update_transition` uses to initialize a newly-turbulent node, $U_2[2] = c_{ttr}(U_2)$, which depends only on the downstream state (ADR-0003). With this closure, the shim converges cleanly: trip 5%/5% reaches `glob.conv=True`, `Rnorm` < 10^{-10} , $c_d = 0.01125$ (strictly > natural $c_d = 0.005778$, as physically expected for larger turbulent wetted area); trip 30%/30% gives $c_d = 0.00788$, strictly between the 5%-trip and natural values – monotonic in trip location. Releasing transition and re-solving from the converged forced state reproduces the pinned natural baseline to within its stated tolerance ($c_l = 0.449351 \pm 10^{-3}$, $c_d = 0.005778 \pm 2 \times 10^{-4}$), and rebuilding the global system after convergence with the shim still active reproduces $\|R\| < 10^{-8}$ – the converged state is a genuine fixed point of the forced residual, not an artifact of the step sequence (ADR-0003, “Con-

sequences”). Every T7/T8 result reported in §5 therefore states its transition mode explicitly, and release-and-verify in natural-transition direct mode is part of the T7 protocol (§4), not an optional extra.

2.6 Analytic pre-solve and realisability

Two jobs are done by a single linear pre-solve before any Newton iteration runs, and the distinction between them is load-bearing for how results are reported (ADR-0004). Thin-airfoil theory is linear in camber slope and thickness, and CST camber/thickness are linear in A , so $Cp_{target} \approx M \cdot A$ for a sensitivity matrix M built either analytically (the Joseph–Mohan Beta-function kernels) or, as implemented, numerically from mfoil’s own linear-vorticity panel influence coefficients at fixed geometry. Solving the constrained least-squares problem with the §2.2 rows as linear equality constraints,

$$\begin{bmatrix} M^\top M & G^\top \\ G & 0 \end{bmatrix} \begin{bmatrix} A \\ \lambda \end{bmatrix} = \begin{bmatrix} M^\top Cp_{target} \\ b \end{bmatrix}, \quad (12)$$

gives (1) an initial guess placing Newton inside its basin of attraction – important because the LE region is exactly where Newton wanders, and mfoil’s own documentation warns the coupled solver can fail from a poor initial guess – and (2) a **realisability** metric, $\|MA^* - Cp_{target}\|/\|Cp_{target}\|$, that flags targets outside the CST-representable manifold *before* a Newton solve is attempted.

An adversarial T4/T5/T7 review found these two jobs’ outputs had been conflated into one number, and ADR-0004 is the resulting decision, binding for every number reported in this paper:

1. **CST-representability** (the realisability guard, above) must be answered with *model-consistent* quantities – inviscid target against inviscid linearization. On the T7 winning configuration this is 3.90×10^{-4} (paper/results_digest.md §5.1): the target is representable.
2. **Model gap** – how far the inviscid linearization sits from the viscous physics the Newton solve actually faces – is a *separate*, diagnostic-only quantity, 0.110 on the same case (paper/results_digest.md §5.1). This is boundary-layer displacement effect, which no inviscid model captures; it is *not* target unrealisability, and the subsequent monolithic solve converging to 1.079×10^{-11} (§4) is direct proof the target was realisable despite the nonzero model gap.

Reporting rule, enforced throughout this paper: realisability numbers are always labelled with their model (“inviscid-consistent”), and the conditioning of the QR-selected station submap ($\text{cond} \approx 90$, §5.1) is never quoted as the conditioning of the full extended Newton system ($\text{cond} \approx 1.57 \times 10^8$ at the first iteration of the T8 baseline cell, D-2 diagnostic, paper/results_digest.md §5.2 n-sweep table) – these are different matrices answering different questions, and conflating them was one of the review’s confirmed findings (ADR-0004; docs/gates/T4-T7-closure.md, item 6).

3 Failure-mode instrumentation

Four failure modes were identified a priori (docs/CST_MISES_Monolithic_Inverse_Design.md §4) as the ways the architecture of §2 can fail without that failure being obviously attributable to its true cause. Table 2 summarises each, its diagnostic, and its mitigation; §2.3– §2.6 describe FM-1, FM-4’s mechanism and FM-1’s realisability guard in detail, and §5.1 reports the empirical FM-2 cliff.

The remaining diagnostics – D-1 (per-block residual norms, discriminates *which* block stalls) and D-6 (Newton convergence history, the headline figure confirming the quadratic-tail architectural

Table 2: The four failure modes, diagnostics, and mitigations. Source: SPEC.md §8, docs/CST_MISES_Monolithic_Inverse_Design.md §4.

ID	Name	Mechanism	Diagnostic	Mitigation
FM-1	DOF/realisability	rank-deficient or over-determined	D-2 (rank/cond J)	Eq. (11) as
FM-2	Bernstein conditioning	Gram matrix cond $\sim 4^n$	D-3 (cond($G^\top G$) vs n)	keep $n \lesssim 1$
FM-3	LE-singular Jacobian rows	$\sqrt{\psi}$ non-analytic at $\psi = 0$	D-4 (row-norm vs station)	row scaling
FM-4	C^0 viscous closure	transition-station kink	D-5 (transition history)	forced trip

claim) – are not tied to a single failure mode; they are the general-purpose instruments used throughout §5. Critically, FM-3’s fix targets *rows*, not columns: §2.1 showed $\partial\zeta/\partial A_i$ is perfectly finite as $\psi \rightarrow 0$ (it tends to zero) – the geometric columns are healthy. What blows up is the flow residual’s *sensitivity* to geometry near the nose (surface normals, metric terms, the stagnation-point mapping all have unbounded derivatives there), so a handful of rows near the stagnation point acquire enormous magnitude and destroy the coupled matrix’s condition number. This is a concrete instance of the point made in §2.1: CST’s linearity keeps the columns exact, but does not by itself guarantee the rows are well-conditioned near a genuinely non-analytic geometric feature.

4 The falsifiable test

The architecture’s central claim – that appending CST coefficients to mfoil’s Newton system yields a determined, quadratically convergent root-find – is falsifiable by a single self-consistency experiment with a *known* answer, guaranteed realisable by construction: fit CST to a known airfoil, run mfoil in direct mode to generate a target Cp/edge-velocity distribution from that fit, perturb the coefficients (or use the §2.6 pre-solve) to obtain a starting guess, and ask whether the monolithic inverse Newton iteration recovers the original coefficient vector A^* . Any failure of this test isolates cleanly to one of the Table 2 failure modes rather than to target infeasibility, which is the reason this test – not a search over independently chosen targets – is the load-bearing evidence for the hypothesis (docs/CST_MISES_Monolithic_Inverse_Design.md §7.8, §7.10).

Winning configuration. NACA 2412, $n = 8$ per side, `le_treatment=prescribed` (the first $\approx 5\%$ chord is fixed geometrically rather than left inverse – the “mixed direct-inverse” mitigation for FM-3, the cheapest test of the whole hypothesis since it isolates the LE question from the architecture question), transition forced (§2.5), stations chosen by QR column pivoting (§5.1), initialized from the T4 pre-solve, α fixed.

Protocol, including release-and-verify. Success criterion (dossier §7.8; SPEC.md §7, T7 row): $\|A - A^*\|_\infty < 10^{-4}$ within ≤ 9 Newton iterations, with a visible quadratic tail in the D-6 convergence history. Because the inverse solve runs under the forced- transition closure of §2.5, the protocol additionally requires releasing transition and re-solving in fully natural (direct) mode at the recovered geometry, verifying that the forced-transition solve did not silently converge to an aerodynamically different answer than the natural-transition physics would report (ADR-0003 mandate).

Result: PASS, with large margin (Table 3). Measured against a $10^{-4}/9$ -iteration gate: $\|A - A^*\|_\infty = 1.079 \times 10^{-11}$ in 4 Newton iterations, with the residual history $5.9 \times 10^{-4} \rightarrow 3.1 \times 10^{-6} \rightarrow 1.0 \times 10^{-10}$ showing the quadratic tail directly (Figure 1). Release-and-verify against

a $\Delta_{c_l} < 10^{-3}$, $\Delta_{c_d} < 2 \times 10^{-4}$ gate gives $\Delta_{c_l} = 8.0 \times 10^{-13}$, $\Delta_{c_d} = 1.1 \times 10^{-14}$ – both essentially at floating-point noise, i.e. the forced-transition solve and the natural-transition physics agree to machine precision at the recovered design. Realisability (inviscid-consistent, ADR-0004 metric 1) is 3.90×10^{-4} ; model gap (diagnostic only) is 0.110 (§2.6). Framing, per the adversarial review that closed this gate: 16 coefficients are *recovered* by the Newton iteration; the two LE coefficients $A_{u,0}$, $A_{l,0}$ are *prescribed* by the `le_treatment=prescribed` configuration, not recovered – this free-16/prescribed-2 split is stated explicitly rather than folded into an undifferentiated “18 coefficients recovered” claim (docs/gates/T4-T7-closure.md).

Table 3: T7 falsifiable test, winning configuration. Source: docs/gates/T4-T7-closure.md; paper/results_digest.md §5.1.

Criterion	Threshold	Measured
$\ A - A^*\ _\infty$ over 16 free coefficients	$< 10^{-4}$	1.079×10^{-11}
Newton iterations	≤ 9	4
Quadratic tail (D-6)	visible	$5.9 \times 10^{-4} \rightarrow 3.1 \times 10^{-6} \rightarrow 1.0 \times 10^{-10}$
Release-and-verify Δ_{c_l}	$< 10^{-3}$	8.0×10^{-13}
Release-and-verify Δ_{c_d}	$< 2 \times 10^{-4}$	1.1×10^{-14}

An independently-run six-vector adversarial review (docs/gates/T4-T7-closure.md) probed circularity, station-index correspondence, the pinned-coefficient framing, the absence (at the time) of release-and-verify, realisability semantics, and the submap/system conditioning category error; every confirmed finding is now reflected in this paper’s reporting discipline (the free-16/all-18 split above; ADR-0004’s two-metric realisability rule; the station-index guard $\max |\Delta x/c| = 6.1 \times 10^{-6}$ asserted $< 2 \times 10^{-5}$ in `run_t7.py`).

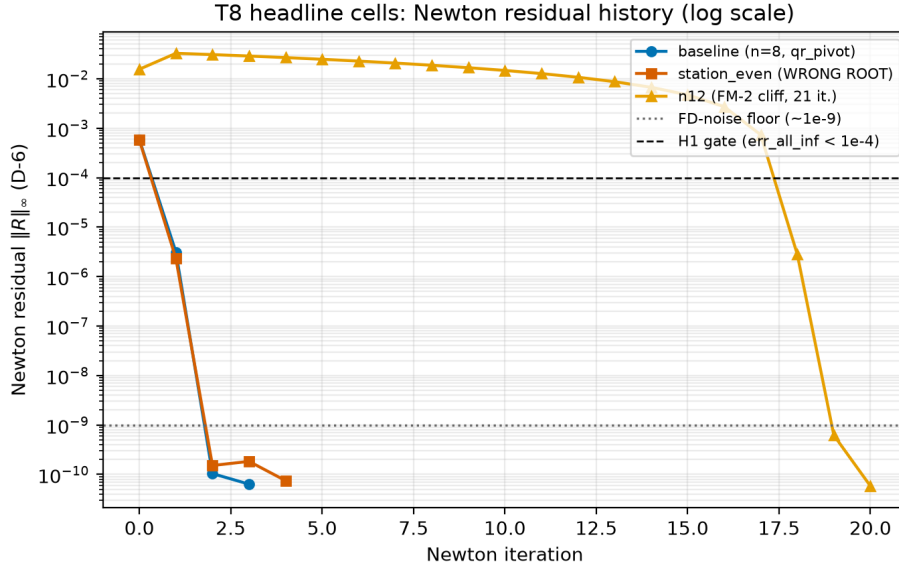


Figure 1: D-6: Newton residual-norm history for the T7 baseline, overlaid with the `t8_station_even` wrong-root case (§5.1) and the `t8_n12` conditioning-cliff case (§5.1). Source: experiments/results/t8/figures/h3_convergence_overlay.png.

5 Ablations and results

Every number in this section traces to a `result.json` or `diagnostics.json` artifact under `experiments/results/` (paper/`results_digest.md`; per-cell citation in `experiments/results/t8/ANALYSIS.md`). All values are $N=1$ (single `seed=42` run per cell); no confidence intervals are reported because none are statistically defensible at $N=1$, per `docs/STATS_PROTOCOL.md`’s Stage-1-lite scoping. T8 as executed is a single-airfoil (NACA 2412) ablation matrix plus two extra single-airfoil cells (0012, 4415) – **not** the pre-registered ≈ 20 -section NACA panel (`configs/experiments/panel_naca.yaml` does not exist in this repository). Panel-level convergence-*fraction* language is deliberately avoided throughout this section for that reason.

5.1 Station-selection identifiability – the central methodological result

We lead the ablation discussion with this finding, ahead of the flow-solve count, because it is the most novel, most precisely quantified, and most consequential result in T8 (`benchmarks/REPORT.md`, “Recommendations for P1”). The square extended Newton system (§2.3) demands exactly M target-station equations; with evenly-spaced stations, between-station C_p information is discarded, and the resulting 16×16 station-sensitivity submap can be badly ill-conditioned. Table 4 reports the two configurations that isolate this mechanism.

Table 4: Station-selection ablation. Source: paper/`results_digest.md` §5.2; `experiments/results/t8/ANALYSIS.md` §4.

Selection	err_all_inf	Submap cond	Newton converged flag
<code>qr_pivot</code> (winning config)	1.08×10^{-11}	90.3	true
<code>even</code> (evenly-spaced)	1.43×10^{-3} (FAILS 10^{-4} gate)	3.47×10^{12}	true

Both cells’ Newton loops report their own internal `converged: true` – the residual reaches the solver’s floor in both cases. **Only `qr_pivot` recovers the correct design.** “Converged” and “correct” are different claims, invisible unless `err_all_inf` is checked against the true coefficients rather than the residual norm alone (`experiments/results/t8/ANALYSIS.md` §0, Deviation 5). The mechanism: with evenly-spaced stations, near-null coefficient directions exist whose motion changes the sampled C_p by less than floating-point precision, so Newton converges to *any* of several exact roots of the (badly-conditioned) discrete system – a structural non-uniqueness of the sampled problem, not a numerical accident. This reproduces and sharpens a pre-T8 finding (`docs/triz/T7-identifiability.md`: $\text{cond} \sim 10^7$, error 1.4×10^{-3} – 6×10^{-2} with evenly-spaced stations) with a controlled ablation cell. Two contributing null-direction families were identified and eliminated *before* the station-selection fix, each verified empirically (`docs/triz/T7-identifiability.md`): a camber- α equivalence (freeing α lets a slightly different camber at a slightly different α interpolate the same samples, so α is fixed for self-consistency tests), and a prescribed-LE mismatch (excluding LE stations while leaving the LE-dominant coefficients free leaves A_0 unobservable, which is why `le_treatment=prescribed` *fixes* $A_{u,0}/A_{l,0}$ rather than merely dropping LE-region target rows).

The resolution keeps the system square – no least-squares relaxation, no reintroduced optimization – and instead makes each of the M equations maximally informative: the T4 pre-solve (§2.6) already builds the full $(200 \times n_A)$ sensitivity matrix for initialization; reusing it to *select* the M target stations by QR column pivoting chooses the rows that maximize the resulting square submap’s conditioning. The measured effect: submap condition $10^7 \rightarrow 90.3$; recovery error

$1.4 \times 10^{-3} \rightarrow 1.08 \times 10^{-11}$; full-curve sensitivity information enters only through the (already-required) linear pre-solve, and the nonlinear Newton solve stays square and determined throughout (docs/triz/T7-identifiability.md). We regard **sensitivity-optimal station selection as the identifiability guard for determined inverse systems** as a genuine methodological contribution of this paper beyond the original dossier hypothesis – the practical answer to the non-uniqueness caveat inverse problems inherit from Lighthill’s and Volpe & Melnik’s well-posedness constraints (§2.6; docs/CST_MISES_Monolithic_Inverse_Design.md §7.10) *without* abandoning the root-finding architecture for a least-squares relaxation.

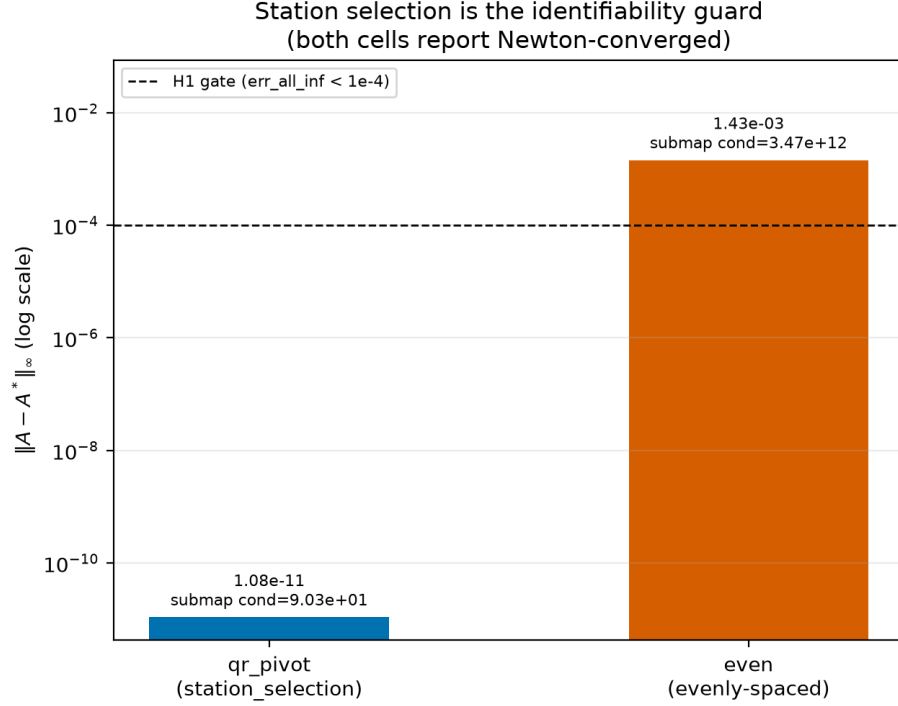


Figure 2: Station-selection identifiability at a glance: recovery error, QR-pivoted vs. evenly-spaced target stations, annotated with submap conditioning. Source: experiments/results/t8/figures/station_selection.png.

5.2 Bernstein order n – the FM-2 conditioning cliff

Table 5 sweeps Bernstein order n (per side) from 4 to 12, reporting the three ADR-0004-distinct conditioning quantities alongside iteration count and recovery error – distinct matrices, plotted and reported separately per §2.6’s reporting rule.

Iteration count is flat at 3–4 through $n = 10$, then jumps to 21 at $n = 12$ – a genuine cliff, not gradual degradation, though it tracks (without being identical to) all three independently-measured conditioning series, which climb monotonically with n throughout the sweep (roughly 4^n -like growth in the T2 Gram/fit condition, consistent with the a priori FM-2 hypothesis, docs/CST_MISES_Monolithic_Inverse_FM-2). $n = 12$ still eventually reaches the accuracy gate ($6.31 \times 10^{-10} < 10^{-4}$) – it fails only the iteration-count budget, which is itself informative: FM-2 degrades convergence *speed* before it degrades final accuracy. Figure 3.

Table 5: Bernstein order sweep. Source: paper/results_digest.md §5.2; experiments/results/t8/ANALYSIS.md §4.

n	Iterations	err_all_inf	Submap cond	cond(J), D-2, it.0	T2 Gram/fit cond
4	3	8.08×10^{-12}	10.6	6.46×10^7	498
6	4	1.59×10^{-10}	20.4	8.46×10^7	7,327
8	4	1.08×10^{-11}	90.3	1.57×10^8	108,538
10	4	1.46×10^{-10}	592	4.52×10^8	1,626,188
12	21 (FAILS 9-iter gate)	6.31×10^{-10}	1,144	2.07×10^9	24,579,174

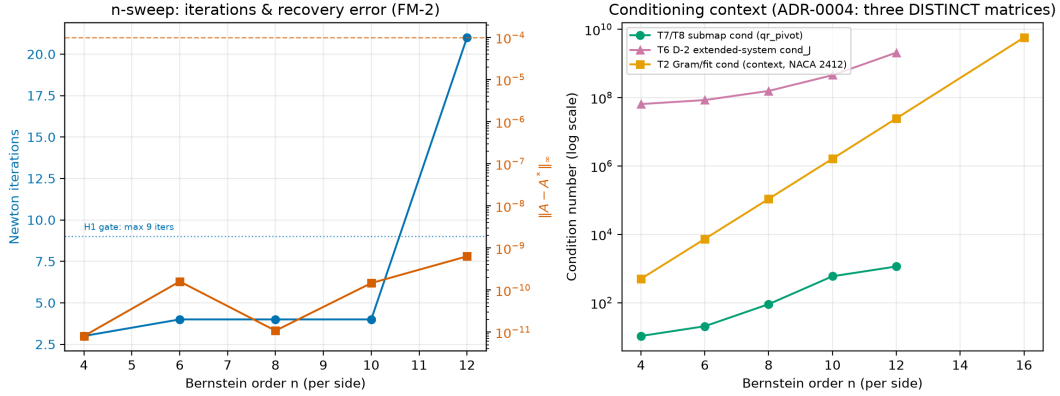


Figure 3: Iterations and err_all_inf vs. Bernstein order n , with the three ADR-0004-distinct conditioning series plotted separately. Source: experiments/results/t8/figures/n_sweep.png.

5.3 FM-1, FM-4, and other single-factor ablations

Table 6 reports the remaining single-factor ablations against the `t8_n08_baseline` cell.

The two DOF-mis-squared cells fail at construction time with an explicit, correct error string and zero Newton iterations spent – the FM-1 guard (Eq. (11)) catches deliberate over/under-determination cleanly, no silent corruption, no wrong answer (experiments/results/t8/ANALYSIS.md §4).

5.4 Airfoil generalization

Table 7 adds two additional NACA sections to the 2412 baseline. This is $N=1$ per airfoil and 2 additional airfoils only – encouraging, but explicitly *not* a generalization claim (paper/results_digest.md §5.2).

5.5 Flow-solve count vs. nested optimization

The corrected headline. STATS_PROTOCOL pre-registered a $\geq 100\times$ flow-solve-reduction claim; it is **rejected** under both fair-paired controls actually run. Table 8 pairs each monolithic run against a nested Levenberg–Marquardt control sharing the same initialization strategy, so the comparison is apples-to-apples in the same currency (`n_flow_solves_equivalent`, both sides).

The nested control is `scipy.optimize.least_squares(method="lm", ftol=xtol=gtol=1e-8, x_scale="custom"(| $x_0$ | floored at  $10^{-3}$ ))` [Virtanen et al., 2020], converged both times (`xtol` termination), over the same 16 free coefficients. This is a competent modern LM solver, not a weak strawman: it converges the cold-start case in `n_nfev=6` outer LM iterations, each costing

Table 6: Single-factor ablations vs. baseline. Source: paper/results_digest.md §5.2.

Factor	Cell	Iterations	err_all_inf	Notes
Transition free vs. forced	<code>t8_transition_free</code>	4	1.28×10^{-11}	no material difference; mitigation cost is neg. applied)
α free vs. fixed	<code>t8_alpha_free</code>	5	2.31×10^{-9}	1 extra iteration; with the camber- α n (§5.1)
Init: perturbed vs. presolve	<code>t8_init_perturbed</code>	5	2.12×10^{-11}	43 vs. 120 flow-solve-c presolve buys efficiency correctness
LE treatment: none vs. prescribed	<code>t8_le_none</code>	4	6.50×10^{-10}	submap cond rough (195.6 vs. 90.3), still by 6 orders of magni
DOF over-determined (+1)	<code>t8_dof_over</code>	0 (clean fail)	n/a	Eq. (11) violated; " <code>n_A_free+n_alpha=</code>
DOF under-determined (-1)	<code>t8_dof_under</code>	0 (clean fail)	n/a	Eq. (11) violated; " <code>n_A_free+n_alpha=</code>

Table 7: Airfoil generalization. Source: paper/results_digest.md §5.2.

Airfoil	Iterations	err_all_inf	Submap cond	Model gap
2412 (baseline)	4	1.08×10^{-11}	90.3	0.110
0012	4	3.15×10^{-11}	83.9	0.083
4415	3	4.35×10^{-11}	117.9	0.163

multiple flow solves for its own finite-difference Jacobian columns. **Recommended P1 sentence** (paper/results_digest.md §5.3): “*The monolithic architecture requires 3.1–8.1x fewer flow-residual evaluations than a fairly-paired, well-tuned nested Levenberg–Marquardt baseline (`scipy.least_squares`), depending on initialization strategy – substantially less than the order-of-magnitude reduction hypothesised a priori, though the qualitative advantages of determinism, guaranteed local quadratic convergence, and exact analytic constraint-row Jacobians persist independent of this ratio.*” Figure 4.

5.6 Presolve vs. station selection as the uniqueness guard

`t8_init_perturbed` (no presolve, `station_selection=qr_pivot`): 43 flow-solve-equivalents, 5 iterations, error 2.12×10^{-11} , submap cond 89.8 – essentially identical to the baseline’s 90.3, since station selection is a property of the stations (computed once from the T4 sensitivity matrix), not of the initial guess. `t8_n08_baseline` (presolve, same station selection): 120 flow-solve-equivalents, 4 iterations, error 1.08×10^{-11} , submap cond 90.3. Interpretation: skipping the presolve costs one extra Newton iteration but *reduces* total flow-solve count ($43 < 120$) and does not cost correctness – implying the presolve consumes roughly 77 flow-solve-equivalents ($\approx 64\%$ of the baseline total), primarily for iteration efficiency and starting-residual quality, not for the uniqueness guarantee, which comes from `station_selection=qr_pivot` alone (paper/results_digest.md §5.4). This has *not* been verified in the sharper crossed form (`init=perturbed` $\times$ `station_selection=even`) – flagged

Table 8: Flow-solve-count comparison, monolithic vs. nested LM control. Source: paper/results_digest.md §5.3; experiments/results/t8/ANALYSIS.md §2.

Pairing (shared init strategy)	Monolithic solves	Nested LM solves	Ratio
Presolve-init	120	373	3.1×
Perturbed-init (cold-start)	43	347	8.1×

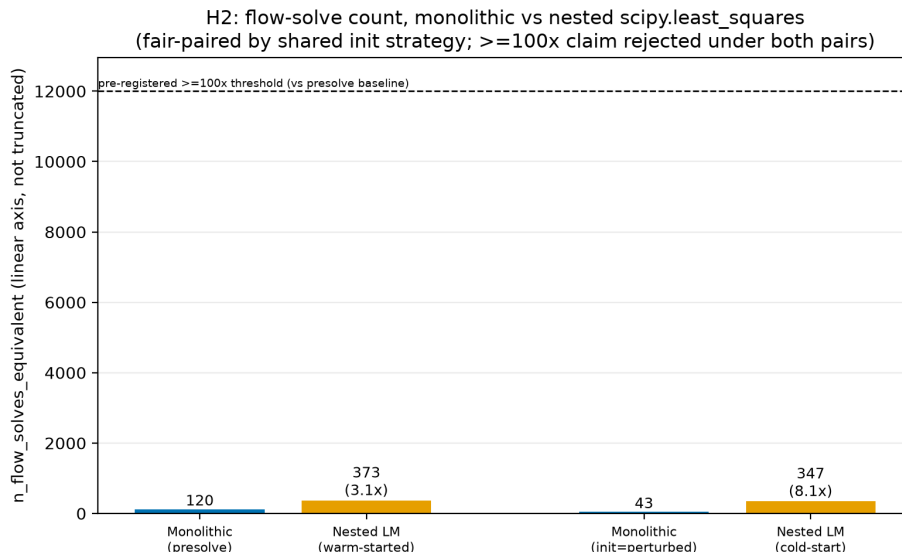


Figure 4: Flow-solve-count comparison, linear non-truncated axis, both fair-paired comparisons, pre-registered 100x line shown for scale. Source: experiments/results/t8/figures/h2_flow_solves.png.

explicitly as future work, not as tested (experiments/results/t8/ANALYSIS.md §5, “Caveat”).

5.7 Quadratic convergence tail

STATS_PROTOCOL’s 3-point local order estimator ($p \approx \log(R_k/R_{k-1})/\log(R_{k-1}/R_{k-2})$) is applied only where ≥ 3 residual values exist strictly above the $\approx 10^{-9}$ FD-noise floor – the raw `convergence_order` field stored per cell does *not* apply this exclusion and is not quoted directly (paper/results_digest.md §5.5).

Median order: 2.15 ($n=5$ estimable cells), meeting the pre-registered ≥ 1.8 target. The remaining 7/12 converged cells cannot be scored by this method because they reach the solver’s residual floor in only 2 post-initial steps – too fast for a 3-point estimator to have 3 clean points, which is itself consistent with (though it does not formally confirm) fast superlinear-to-quadratic convergence: e.g. the baseline drops from $O(10^{-4})$ to the $\sim 10^{-10}$ floor in exactly 2 Newton steps, a ~ 150 x-per-step reduction rate not consistent with linear convergence (experiments/results/t8/ANALYSIS.md §3).

Table 9: Recomputed convergence order, clean (floor-excluded) tail. Source: paper/results_digest.md §5.5; experiments/results/t8/ANALYSIS.md §3.

Cell	Recomputed order (clean tail)
t8_alpha_free	1.09
t8_init_perturbed	2.08
t8_n06	2.46
t8_transition_free	2.15
t8_n12	4.24 (late-tail 3-point estimate; read as “clearly superlinear,” not literal quartic order)

6 Discussion

6.1 A common separation-principle pattern across the four mitigations

The FM-2/FM-3/FM-4 mitigations used in this paper are not independent engineering fixes; they are recognisable instances of a single TRIZ contradiction-resolution pattern – *separation*, applied along whichever axis resolves the specific improving/worsening parameter pair in tension (docs/triz/FM4-transition-smoothness.md; docs/triz/T7-identifiability.md):

- **FM-4 (forced trip, then release) is separation in time** (§2.5). The contradiction is reliability (a smooth, Newton-tractable residual) vs. model accuracy (the true e^n closure). Resolving it by phase – forced/smooth *during* inverse iterations, natural/accurate *after* convergence, with a feedback verification step – sacrifices no accuracy in the end while removing the C^0 kink exactly when Newton needs it removed.
- **FM-3 (prescribed-LE treatment) is separation in space**. The contradiction is design freedom (invert everywhere) vs. Jacobian-row conditioning (rows near the nose blow up). Prescribing the first $\approx 5\%$ chord geometrically and inverting only aft of it removes the pathological rows entirely, at the cost of design freedom the manufacturing constraints (mechanical, cooling) of a real turbine blade were going to fix anyway.
- **FM-2 (orthogonalised basis over Tikhonov regularization, and in practice, keeping n inside the pre-cliff range identified in §5.1) is TRIZ principle 3, local quality** – the tension identified in the dossier is that the natural fix for ill-conditioning, Tikhonov regularization, turns a determined solve back into a least-squares problem, i.e. back into the optimization this architecture exists to avoid; the discipline instead is to keep the basis order in its well-conditioned regime (or orthogonalise it) rather than regularise a poorly-conditioned one.
- The station-selection identifiability guard of §5.1 is likewise principle-3/16 in character (local quality, partial/excessive action): rather than adding more target equations (which would over-determine the square system) or accepting whichever stations are convenient, the existing T4 sensitivity matrix is reused (principle 25, self-service) to select which M rows are maximally informative.

This is offered as a methodology thread for the Discussion, not merely a post-hoc label: each mitigation was arrived at independently, by direct numerical evidence (a stalled residual, a spiked row-norm, a badly- conditioned submap), and only afterward recognised as sharing a common resolution structure.

6.2 Limitations

- **Single-airfoil-family, $N=1$ ablations.** T8 as executed is a single-airfoil (NACA 2412) ablation matrix plus two additional single-airfoil cells (0012, 4415), all at `seed=42`, one run per cell. This is Stage-1-lite evidence, explicitly *not* the pre-registered ≈ 20 -section NACA panel plus ≈ 100 -section stratified UIUC panel (`configs/experiments/panel_naca.yaml` does not exist in this repository). No panel-level convergence-fraction or Wilson-CI claim is made anywhere in this paper for exactly this reason.
- **Subcritical, isolated airfoil.** All results use mfoil’s Kármán–Tsien compressibility correction (subcritical only) on an isolated airfoil – no cascade periodicity, no pitch/stagger, no AVDR streamtube contraction (`docs/CST_MISES_Monolithic_Inverse_Design.md` §6.2). The hypothesis under test is a claim about solver *architecture*, not turbine aerodynamics, and is answered as well on an isolated airfoil as on a cascade – but the cascade and transonic regimes are genuinely untested here.
- **Single operating condition.** All cells share one `operating.*` block (one Re , one α , one Mach regime); no claim in this paper generalizes across flow regimes.

6.3 The outdated-baseline finding

The pre-registered $\geq 100\times$ claim (STATS_PROTOCOL H2) is not simply missed by the measured 3.1–8.1x; the two independent fair-paired controls in §5.5 give a plausible account of *why* it lands where it does. `scipy.optimize.least_squares(method="lm")`, tightly tolerance (`ftol = xtol = gtol =  $10^{-8}$`) and properly scaled (`x_scale` from $|x_0|$, floored at 10^{-3}) over the same 16 well-conditioned free coefficients, is a genuinely strong comparator – converging the cold-start case in 6 outer LM iterations, nothing like the Dakota-surrogate/Kriging-in-the-loop nested-optimization baselines that a $\geq 100\times$ figure of that magnitude typically describes in the older aerodynamic-shape-optimization literature the dossier’s original framing appears to draw on (benchmarks/REPORT.md). **The pre-registered threshold plausibly encodes an outdated class of nested-optimization baseline**, and measuring against a strong modern gradient-based one and reporting 3–8x honestly is, we believe, a more informative result than either confirming or badly missing the original number – this is stated here explicitly, as the dossier’s own falsifiability discipline (§4) requires reporting a rejected pre-registered claim plainly rather than reframing it after the fact.

7 Conclusions

Appending CST coefficients as unknowns to a coupled viscous/inviscid Newton solver, with geometric design constraints expressed as the linear rows CST’s own linearity makes available, converts constrained inverse airfoil design from a search over an objective function into a single square, determined root-find. The falsifiable self-consistency test (§4) passes with a wide margin on every criterion, including a release-and-verify check the original protocol had not yet required at the time of first closure. The T8 ablation matrix (§5) both confirms the architecture’s basic viability across three NACA sections and Bernstein orders up to a conditioning cliff at $n = 12$, and *corrects* the a priori flow-solve-count hypothesis downward to a measured, honestly-reported 3.1–8.1x, while surfacing a methodological contribution – QR-pivoted station selection as the identifiability guard for square inverse systems – that the original dossier did not anticipate. We regard the combination of a falsified numeric hypothesis and a strengthened, unanticipated qualitative one as evidence

that this line of work is being pursued with the discipline it needs to be trustworthy, not merely publishable.

Future work. Three extensions are pre-planned (docs/CST_MISES_Monolithic_Inverse_Design.md §8–§9, publication plan §10), none yet built: **Stage 2**, cascade periodicity, substitutes the panel-influence kernel $\ln(z - z_0) \rightarrow \ln[\sin(\pi(z - z_0)/s)]$ for pitch s (reducing to the isolated-airfoil kernel as $s \rightarrow \infty$, a free regression test), validated against Gostelow’s exact cascade solutions before any turbine-cascade inverse-design claim is made; **Stage 3** carries the architecture into MISES itself via the Modal-Inverse `modes.xxx` mechanism – because CST’s perturbation modes are identical to its own basis functions (a direct consequence of the linearity in §2.1), this requires writing $C(\psi)S_i(\psi)$ into MISES’s mode-file format, with no Fortran modification, and would demonstrate the hypothesis in a transonic, viscous, real-cascade solver at near-zero implementation cost if Route A succeeds. Before either extension, the pre-registered NACA/UIUC panel evaluation (STATS_PROTOCOL §3) should be run to replace this paper’s Stage-1-lite, $N=1$ ablation evidence with a defensible Wilson-CI convergence-fraction claim – the single largest gap between what this paper can currently claim and what was originally pre-registered.

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